Blind Source Separation: A Review and Analysis

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***Abstract*— Blind Source Separation (BSS) refers to a problem where both the sources and the mixing methodology are unknown, only mixture signals are available for further separation process. In several situations it is desirable to recover all individual sources from the mixed signal, or at least to segregate a particular source. In laboratory condition, most of the algorithms works very fine where input signals, no. of source present in the mixture, mixing methodology etc are well known to the separation process. But in real-life scenario the problem is much more complicated and it begins with the input signal, a mixture where most of the parameters are unknown. This paper will try to summarize those approaches taken previously to solve this problem and an experiment of source separation which will mix using Independent Component Analysis (ICA) and then de-mix those source signals using ICA as the basic/prime approach.**

Keywords—Blind Source Separation; Independent Component Analysis; Mixing; De-mixing

I. INTRODUCTION

A class of unsupervised methods has been developed and applied to audio signal processing. These methods, usually referred to as BSS are based on the sole assumption that the sources are mutually independent. BSS algorithms pursue a similar goal as beam-forming: to reduce interferences. A major difference to beam-forming is that BSS does not require any information about the target position, and BSS is sometimes termed "blind beam-forming" [1]. Another difference is that BSS recovers all individual sources simultaneously from the mixture, while beam-forming concentrates extracting only one target source.

Audio separation technique has found its place in numerous practical applications, such as: voice recognition systems [2], improving hearing aid technology [3], etc.

Several Approaches are there like

- Binary mask for separation [4]
- Independent Component Analysis (ICA) method [5]
- Time-Frequency masking approach for speech separation [6]
- The Degenerate Un-mixing Estimation Technique (DUET) algorithm [7]
- Adaptive Blind Source Separation [8]
- Online Speech Source Separation [9]
- BSS in image separation [10, 11]
- Clustering and MMSE-based filtering [12]
- Similarity Measures [13]
- Independence [13]
- Non-negativity [13]
- Sparseness [13]
- Temporal Continuity [13]
- Autoregressive Modeling [14]

Some recent research Trends are –

- Blind MIMO de-convolution of any Source distributions via High-Order Spectra [15]
- Autoregressive Modeling of Temporal Envelopes[16]
- An Expectation–Maximization Method for Spatio-Temporal Blind Source Separation Using an AR-MOG Source Model [17]
- Second Order statistics and modulation codes [18]
- Adaptive Semi-blind Signal Separation Approach Using Temporal Structure of Sources [19]
- IEEE-Blind Source Separation Based on Signal Temporal Predictability [20]

Although many different source separation algorithms are available, the following fundamental approaches can summarize their principles (see Fig. 1[21]).

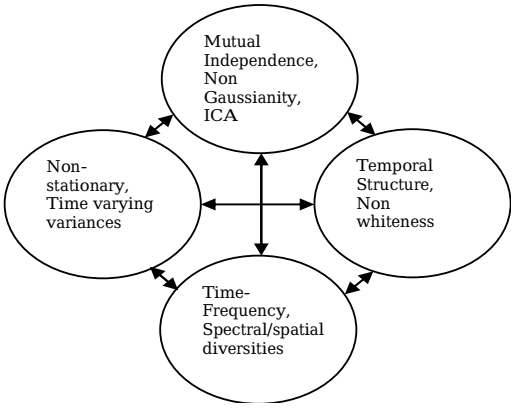


Fig. 1: Basic Approaches

II. MOTIVATION AND OBJECTIVE OF THE WORK

The main motivation behind this work - problem of cocktail party situation, noise separation from automatic dialog systems for in-vehicle navigation assistance systems etc.

The cocktail party problem is a classic example demonstrating the need for robust audio separation algorithms. Imagine being

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in a loud party, trying to talk with a friend. During your conversation the sounds that reach your ears are a complicated mix of music, other people talking, glasses tinkling and so on. Even though all of these sounds arrive at the ear as a single waveform, still you are able to understand your friend's voice from the mixture and still have time to enjoy the music and keep alert of your surroundings by audibly checking the scene. Machines, on the other hand, get confused when even a distant and faint sound such as a passing car goes by as you talk to your microphone.

Speech is natural and therefore a privileged communication modality for humans. For example in cars, safety and convenience issues require hands free (or "seamless") speech-based human–machine interfaces for the driver to manipulate complex functionalities and devices while driving. Applications include hands-free phone calls as well as more advanced functions such as automatic dialog systems for in-vehicle navigation assistance systems. With a seamless speech input, such interfaces increase comfort but have to face several issues-

i) The signal-to-noise ratio (SNR) at a given microphone can be weak relative to the background noise since the signal energy is inversely proportional to the square of the distance to the sound source. Moreover, room acoustics leads to a reverberated speech signal.

ii) Interferences, such as speech from the co-driver, may greatly hamper the speech recognizer performance, which is crucial for human–machine dialog applications. Separation of the target speaker during periods of competing speech from the co-driver represents a particular challenge. This is because the characteristics of the interferer signals cannot be directly estimated from the microphone signals during these periods. This problem is of particular importance since spontaneous multiparty speech contains lots of overlaps between the speech flows of the participants.

These issues make the seamless speech input a challenging problem. Before recognizing speech as a sequence of words, an important preprocessing step is to de-noise the speech signal from its perturbations.

The problem of source separation in the auditory domain has received a lot of attention and numerous approaches have been developed. In general these approaches are divided into two categories - single input and multi input algorithms [22] i.e. Single Input Approaches and Multi Input Approaches. Fig. 2 and Fig. 3[23] show the single channel input approach and multi-channel input approach respectively. In Fig. 3 various types of source input signals before multi-channel mixing can be seen. In our experiment we are using Multi-Channel Input Approach.

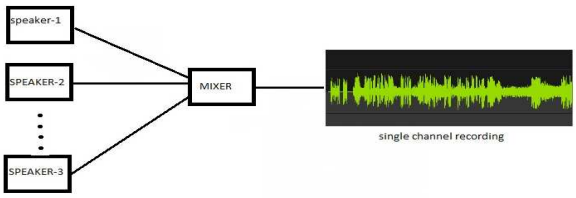


Fig. 2 Single Channel Mixing

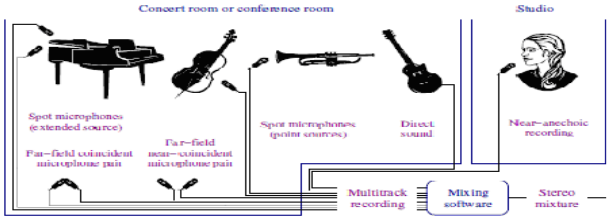


Fig. 3 Multi-channel input Mixing

The main objective of this work is to distinguish the individual signals present in an unknown mixed signal which may contain signals from various sources- a perfect example of this kind of mixed signals are cocktail party.

Basically, a mixed signal can be made of single channel or multi-channel signals. Single channel means mixed signal is created using single microphone connected with single channel where as multi-channel mixture signal is created using more than one microphone connected with more than one channels. Separating multiple sources from a single channel mixed source is more difficult than separating multiple sources from a multi-channel mixed source. So we first try to concentrate on the easier separation from multi-channel mixer then we will proceed to separate single channel mixer.

It may also be observed that in the mixed signals some of the signals are low i.e. amplitude is very low, it happens because the signal itself is of low amplitude or it may travel long distance before get recorded, so in such case when the post processing is done on that signal, the low amplitude get most affected. In such cases, signal normalization is very much essential so that the mixed signal does not get affected at the time of processing.

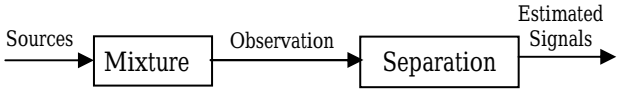


Fig. 4 Experimental Diagram

Fig. 4 shows bunch of input signal(source signals) enters into the Mixture block for mixing and then using optimization algorithm de-mixer separates the input signals in Separation block according to their source and we get the output signals(Estimated signals).

System will perform the task of Blind Source Separation (BSS) very efficiently if we can solve issues regarding BSS such as:

i) How many channels are used for data recording?
ii) What is the number of the speakers in the mixed speech?
iii) Background noise etc.

Depending on these conditions, sources have to be separated with least degradation of the individual speech. Sometimes it also happens that the amount of speech i.e. the duration of a particular speaker speech is so small that it is quit impossible to segregate the speech. Considering all such constraints BSS is quite difficult task to perform. Still through our experiment, the main objective will be to develop a model, which will perform well to segregate speech signals from mixed signals.

III. System Overview

Fig. 5 shows the basic block diagram and also system flow chart of the proposed system. There are mainly five main blocks namely: Pre-processing, Mixing, Generation of Mixing coefficient matrix, De-mixing and Binary Mask Creation.

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A. Pre-processing Stage

In Pre-processing stage source signals are segmented up to the minimum duration of the two source signals. Fundamental frequency is taken as 22.5 KHz and 16 bit Mono.

B. Mixing

After Pre-processing stage required source signals are obtained. Now using basic mixing rule of ICA as shown in equation (1) mixed signals are obtained.

$$X = A * S \tag{1}$$

Where 'X' is the mixed signal matrix, 'S' is the Source signal matrix and 'A' is Mixing Coefficient Matrix.

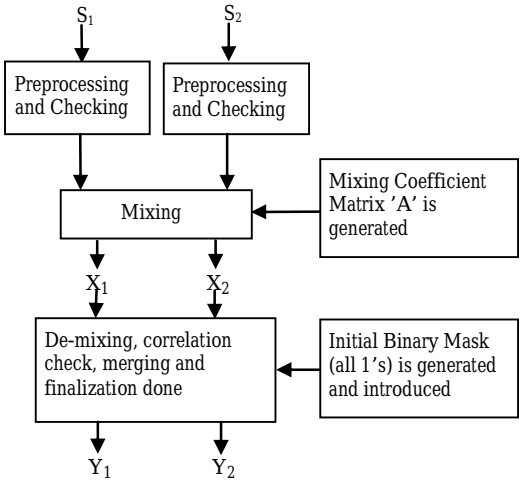


Fig. 5 Overall Block diagram as well as system flow chart

C. Generation of Mixing coefficient matrix

Mixing Coefficient Matrix 'A' of order N x N is generated using the formula 1+rcosθ, 1-rcosθ, where θ is randomly chosen between 0 to 180 degrees and gain r is 0.5 and N is number of source signal. In our case N=2.

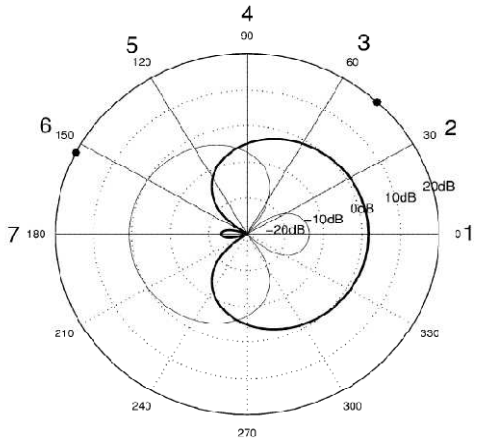


Fig. 6 Selection of random sensor location

D. De-mixing and Binary Mask Creation

Mixed signals (in our case no. of mixed signal is 2 as we are working on determined condition of ICA, so no. of mixed signals = no. of source signals) and Binary Mask (initially taken all 1's) enter in the de-mixing stage where mixed signals are segregated using the formula

$$Y = W * X \tag{2}$$

where W= Inverse of A , After each iteration new Binary Mask is generated according to the mixed signal, what type of windowing (in our case Triangular) is used and what DFT length (in our case 2048) is chosen;

$$BM_1 = \begin{cases} 1, & if\ (|Y1| > \tau(|Y2|)) \\ 0, & otherwise \end{cases} \tag{3}$$

$$BM_2 = \begin{cases} 1, & if\ (|Y2| > \tau(|Y1|)) \\ 0, & otherwise \end{cases} \tag{4}$$

Here τ is a parameter in equation (3) and equation (4) [24] which control how sparse the masks should be, i.e., how much of the interfering signals can be removed in each iteration. If, the two estimated masks together contain the same number of retained T–F units (i.e., equal to 1) as the previous mask; from that new intermediate segregated signals Y is generated [24]. Y may or may not be the actual desired signal. In each iterations the power of the signal is checked if it is above the threshold value (or minimum value of the given source signals power is below threshold value) Correlation is checked between the signals in Y and the masks. If there exists any correlation that mask is merged with intermediate signal and re-estimation is performed again on that signal when there is no correlation present between the intermediate signals and the masks i.e. Gaussian signals are transformed into non-Gaussian signals.

After assuring that there is no correlation present, finalization process is done on the intermediate signals and final output signals are written on the memory.

IV. Experiment Details

Essential requirement to perform Blind Source Separation (BSS) are:

A. Recording material preparation

To perform this Source Separation experiment basic requirement is speech signal i.e. what should be the speech signal like (male/female)? For that recording of speech signal is required and subsequently the content of the recording material also have to be finalized. After finalization of the reading materials, one more important thing is proper informant selection depending on various attributes like pitch, voice quality, clarity, etc. with the help of the selected informant speech data are recorded in a controlled studio environment.

B. Recording set- up

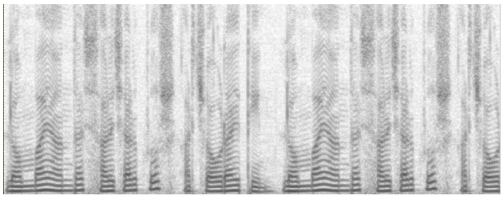
A well equipped studio is required for a good quality of speech data recording.

- Microphones used: - Shure Dynamic Mic
- 3 channel mixer is used
- Sound Card - Creative sound blaster
- S/w used for recording is cool edit pro 2000

Speech data recording format: 22.5 KHz 16-bit Mono

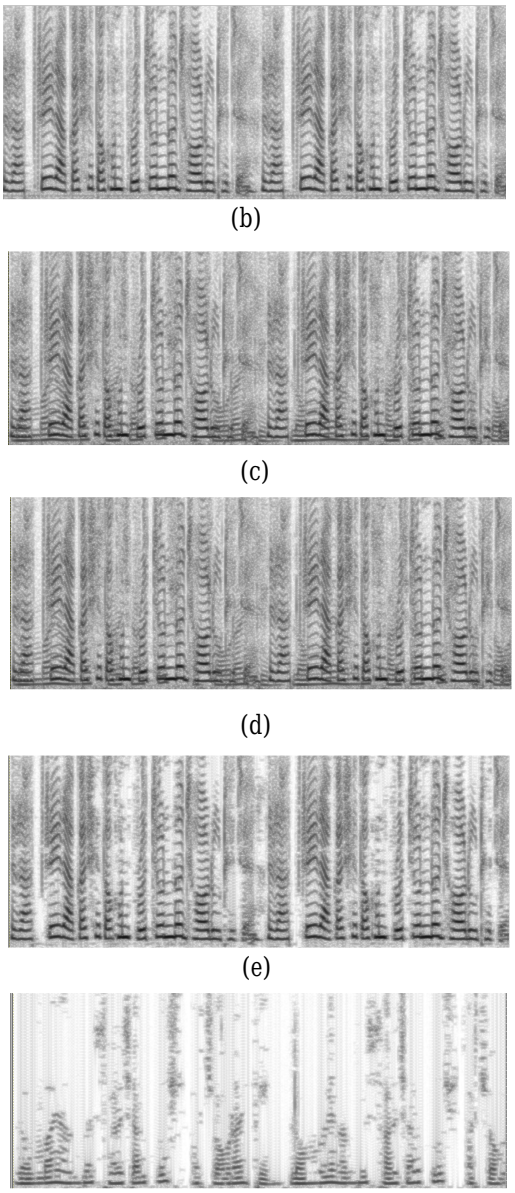
E. Preparation of the sample data set used in the experiment

According to the requirement of the experiment required length signals are augmented from the original recorded speech signals. This augmented section is used as an input of the program of Source Separation.



(a)

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(b) (c) (d) (e) (f)

Fig. 7 (a) and (b) shows the spectrographic view of the input source signals, (c) and (d) shows the spectrographic view of the mixed signals, (e) and (f) shows the spectrographic view of the estimated output signals.

Here the estimated output signal 1 is the estimation of the input source signal 2 and the estimated output signal 2 is the estimation of the input source signal 1.

V. Observations

In this report, a comprehensive study has been done with respect to the various aspect of source separation: – multi-source multi-channel and multi-source single channel. The well known techniques like Adaptive source separation, echo cancellation, Grouping Separated Frequency Components, Bayesian Approach, Filtering techniques, Multi-channel identification based on sparsity, Single-channel identification based on advanced models, Hybrid identification techniques Conclusion and promising approaches etc. are finding their usability in many such source separation fields. Source separation is also widely use in image processing. We believe that the field will experience a paradigm shift in the foreseeable future with the focus being more on application oriented, domain specific work generating considerable impact in day to-day life. We have compiled research trends in Blind Signal separation using Google search tool and citation scores. We have discussed the approaches, new ideas and sub problems emerging out of core BSS in the recent years. Meanwhile we do believe that the quest for robust and reliable BSS understanding technology should continue [25].

The BSS problem remains far from being solved. Realistic audio mixtures are better separated by DUET-like methods or hybrid models. However the quality remains insufficient for demanding applications (hearing aids, karaoke, etc.).
Several issues account for this:

- Reverberation,
- Time-frequency overlap of the sources,
- Possibly source movements, similarities between source characteristics or proximity of source directions.

BSS methods have been proposed independently by researchers from different communities (statistical signal processing, audio signal processing, cognitive psychology) since the early 90's. From the above discussion it is clear the issues which are required to consider most for the understanding of the source separation are like- make clear the modeling assumptions and performance limitations of each approach, emphasize the importance of audio-specific issues in the design of BSS algorithms.
After analyzing the estimated output signals (shown above), it is observed that there are some mismatch between the actual input signal and the estimated output signal. Moreover, quality of both of the signals i.e. input and estimated output is nearly same but signal quality is deteriorated.

It is also observed that in the estimated output signal some of the frequency bands are deteriorated or missing which were present in the original input source signal. Phase mismatch (phase lead or phase lag) is also observed.
And of course it is visible that one of the input signals is dominating in the mixed signal (here source 2 signal is dominating over source 1 signal in the mixed signal).

VI. Perceptual Evaluation

In order to test the perceptual appropriateness of the estimated source signals a perceptual experiment is conducted for testing the functional equivalence of the estimated source signals with the original source signals. Altogether 10 stimuli signals are created by using the experiment on ICA on the original sources which first mixed then estimated by ICA again. For this evaluation 5 subjects, 3 male (L1, L2, L5) and 2 female (L3, L4), are selected, their age ranging from 20 to 30. All subjects are native speakers of Standard Colloquial Bangla and four of them are speech experts.
In this experiment the 20 estimated and original '.wav' signals are presented to the subjects for listening test. The original and estimated sentences are randomly mixed up and presented to the subjects for giving their judgment in 5 point score (Excellent=5, Very Good=4, Good=3, Average=2 and Bad=1). Table 1 represents the tabulated mean opinion scores for all signals of each subject for original as well as estimated signals. It is observed that the difference of the mean opinion score of the subjects for the estimated signals is nearly close to the original signals. In the case of non speech expert (L2) the grade difference is 0.1 which is very minute and for speech experts (L1, L3, L4, and L5) it is less than 1.

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TABLE I RESULT OF LISTENING TEST

		L1	L2	L3	L4	L5
Original Signals	***Avg***	5	4.6	4.3	4.3	4.1
	SD	0	0.516	0.258	0.394	0.738
Estimated Signals	***Avg***	4.4	4.5	3.84	3.2	3.3
	SD	0.516	0.527	0.615	0.483	0.675

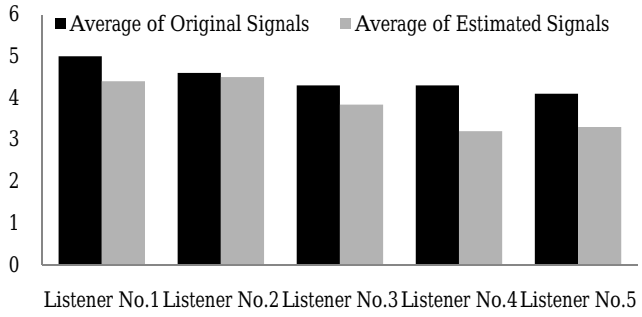


Fig.8 Bar chart of the Listening Test

The average score for the original signals is 4.32 and for the estimated signals it is 4.088. The average grade difference between the original and estimated signals has been found to be less than one grade which is quite encouraging result. Fig. 8 graphically represents the mean opinion score to visualize the closeness of the employed prosodic rules to the original sentences.

VII. LIMITATIONS AND FUTURE SCOPE

Despite success achieved by this method in various fields of BSS, few limitations still resides. Some of them are-

- Number of Sources must be predefined.
- All sources need to have same fundamental frequency.
- Number of source signals must be less than or equal to number of mixed signals.
- Works properly for controlled environment (studio) mixing only.
- Desired result is not achieved in presence of noise in mixing process.
- Mixing Coefficient Matrix 'A' doesn't change its values enormously.

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VIII. REFERENCES

[1] J.F. Cardoso and A. Souloumiac. Blind beamforming for non-Gaussian signals. Proceedings of the IEEE, Vol. 140, No. 6, pp. 362–370, December 1993.

[2] S.Low, R.Togneri, and S.Nordholm, "Spatio-temporal processing for distant speech recognition," in IEEE International Conference on Acoustics, Speech and Signal Processing,, 'pp. 100 1-1004, (2004).

[3] J. Anemuller, "Adaptive separation of acoustic sources for anechoic conditions: A constrained frequency domain approach," Speech Communication, vol. 39, no. 1-2, pp. 79-95, (2003).

[4] D. Ayll´on , R. Gil-Pita, P. Jarabo-Amores, M. Rosa-Zurera, C. LLerena-Aguilar "Energy Weighted Mean Shift Algorithm For Speech Source Separation" 2011 IEEE Statistical Signal Processing Workshop (SSP).

[5] P. Comon, "Independent component analysis, a new concept," Signal Processing, vol. 36, no. 3, pp. 287–314, 1994.

[6] D.Wang, "Time-Frequency Masking for Speech Separation and Its Potential for Hearing Aid Design," Trends in Amplification, vol. 12, no. 4, pp. 332–353, 2008.

[7] O.Yilmaz and S.Rickard,"Blind separation of speech mixtures via time-frequency masking," IEEE Transactions on Signal Processing, vol. 52, no. 7, pp. 1830–1847,2004.

[8] Huan Tao; Jian-yun Zhang; Lin Yu; "Adaptive Blind Source Seperation Using Temporal Predictability," Communications, Circuits and Systems Proceedings, 2006 International Conference on , vol.1, no., pp.280-283, 25-28 June 2006.

[9] Togami, M.; "Online speech source separation based on maximum likelihood of local Gaussian modeling," Acoustics, Speech and Signal Processing (ICASSP), 2011 IEEE International Conference on , vol., no., pp.213-216, 22-27 May 2011.

[10] Sarit Shwartz, Yoav Y Schechner, Michael Zibulevsky, "Efficient separation of Convolutive Image Mixtures", Proc. ICA . Conf. On ICA and Blind Source Separation LNCs 3 889,pp 246-253,(2006).

[11] Konstantinos, Diamantaras Theophilos Papadimitriou, "Blind Separation of Reflections using the Image Mixtures Ratio", IEEE, (2005).

[12] Mehrez Souden, Shoko Araki, Keisuke Kinoshita, Tomohiro Nakatani, Hiroshi Sawada , "Simultaneous Speech Source Separation And Noise Reduction via Clustering and MMSE-Based Filtering", NTT Communication Science Laboratories Signal Processing Research Group, Kyoto, Japan.

[13] Matthew J. Crossley, "Methods of Single-Channel Music Source Separation", June 13, 2010.

[14] A. El-Jaroudi and J. Makhoul, "Discrete all-pole modeling", IEEE Trans. Signal Process., vol. 39, pp. 411–423, 1991.

[15] M. Taoufiki, A. Adib, D. Aboutajdine, s. Biaz ; "Blind MIMO Deconvolution of any Source Distributions via High-Order Spectra"IEEE-978-1-4244-2703-1/08/$25.00; 2008.

[16] Marios Athineos, Student Member, IEEE, and Daniel P. W. Ellis, Senior Member, IEEE; "Autoregressive Modeling of Temporal Envelopes"; IEEE transactions on signal processing, vol. 55, no. 11, November 2007

[17] Kenneth E. Hild II, Senior Member, IEEE, Hagai T. Attias, and Srikantan S. Nagarajan, Senior Member,IEEE ;IEEE TRANSACTIONS ON NEURAL NETWORKS, "An Expectation–Maximization Spatio–Temporal Blind Source Separation Using an AR-MOG Source Model"; VOL. 19, NO.3, MARCH -2008.

[18] Carlos Alexandre R. FERNANDES, G´erard FAVIER,I3S Laboratory,CNRS-University of Nice-Sophia Antipolis, 2000 route des Lucioles, BP 121,06903, Sophia- Antipolis Cedex, France.acarlos@i3s.unice.fr and favier@i3s.unice.fr and Jo˜ao Cesar M. MOTA, GTEL Laboratory, Federal University of Cear´a,Campus do Pici, 60.755-640,6007 Fortaleza, Brazil.mota@gtel.ufc. "Blind Source Separation and identification of nonlinear multiuser channels using Second Order statistics and Modulation Codes"; IEEE; 2007.

[19] Rui Li, School of Sciences, Henan University of Technology, Zhengzhou 450052, P.R.China, Department of Mathematics, Zhengzhou University, Zhengzhou, 450000, P.R.China, slxlr@haut.edu.cn, Fasong Wang, China Electronics Technology Group Corporation the 27th Research Institute, Zhengzhou 450005, P.R. China, fasongwang@126.com; "Adaptive Semiblind Signal Separation Approach Using Temporal Structure of Sources" , 2007.

[20] Z. S. He, S. L. Xie, S. X. Ding, and A. Cichocki, "Convolutive blind source separation in the frequency domain based on sparse representation", IEEE Trans. Audio, Speech, and Language Process, vol. 15, no. 5, pp. 1551-1563, 2007.

[21] Sangeeta D. Jadhav, Anj all S. Bhalchandra, "Blind Source Separation: Trends of New Age -a Review".

[22] Paris J. Smaragdis "Information Theoretic Approaches to Source Separation".

[23] Emmanuel Vincent, "Blind Audio Source Separation A review of state-of-the-art techniques", Centre for Digital Music Electronic Engineering Department Queen Mary, University of London.

[24] Two-Microphone Separation of Speech Mixtures Michael Syskind Pedersen, Member, IEEE, DeLiang Wang, Fellow, IEEE, Jan Larsen, Senior Member, IEEE, and Ulrik Kjems, Member, IEEE.

[25] Jadhav, S.D.; Bhalchandra, A.S., "Blind source separation: trends of new age – a review," Wireless, Mobile and Multimedia Networks, 2008. IET International Conference on , vol., no., pp.251-254, 11-12 Jan. 2008.

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